C3.2 Energetics

Summary
Chemical reactions are accompanied by an energy change. A simple model involving the breaking and making of chemical bonds can be used to interpret and calculate the energy change.

Underlying knowledge and understanding
Learners should be familiar with exothermic and endothermic chemical reactions.

Common misconceptions
Learners commonly have the idea that energy is lost or used up. They do not grasp the idea that energy is transferred. Learners also wrongly think that energy is released when bonds break and do not link this release of energy with the formation of bonds. They also may think for example that a candle burning is endothermic because heat is needed to initiate the reaction.

Tiering
Statements shown in **bold** type will only be tested in the Higher Tier papers.
All other statements will be assessed in both Foundation and Higher Tier papers.

Reference	Mathematical learning outcomes	Mathematical skills
CM3.2i	interpretation of charts and graphs when dealing with reaction profiles	M4a
CM3.2ii	arithmetic computation when calculating energy changes	M1a

Topic content		Opportunities to cover:		Practical suggestions
Learning outcomes	To include	Maths	Working scientifically	
C3.2a distinguish between endothermic and exothermic reactions on the basis of the temperature change of the surroundings			WS1.4c	Measuring the temperature change in reactions. (PAG C8)
C3.2b draw and label a reaction profile for an exothermic and an endothermic reaction	activation energy, energy change, reactants and products	M4a	WS1.3b, WS1.3c, WS1.3d, WS1.3e, WS1.3g, WS1.3h, WS1.4c	
C3.2c explain activation energy as the energy needed for a reaction to occur			WS1.4c	
C3.2d calculate energy changes in a chemical reaction by considering bond making and bond breaking energies		M1a	WS1.3c, WS1.4c	